

Ride Hailing App Loyalty Program

User Segmentation & Churn Risk Analysis

Maximizing marketing ROI and preserving high-value customer lifetime value through data-driven RFM behavioral tiering and explainable predictive risk-scoring.



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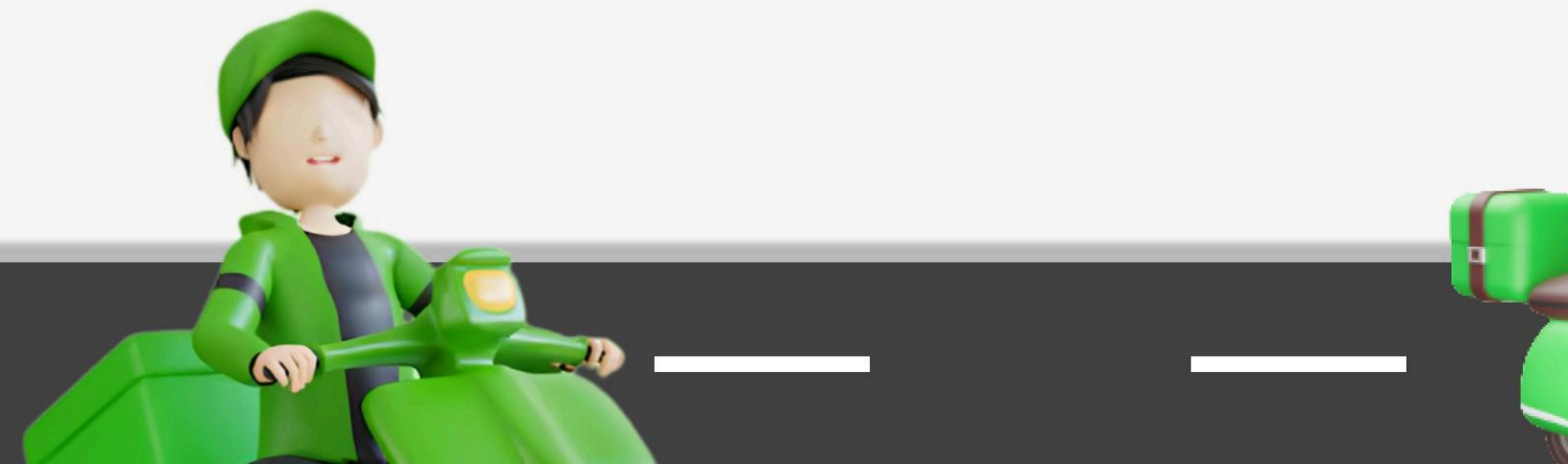
Background

The **Ride Hailing Loyalty Program** an OVO points-based ecosystem covering GrabBike, GrabCar, GrabFood, GrabMart, and GrabExpress is a **critical user retention tool** in Indonesia.

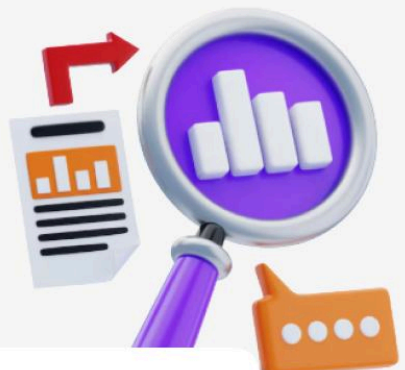
Operating in a **highly competitive market** with **zero switching costs**, user retention depends heavily on **perceived value** rather than habit alone. Well designed tier and points based programs **increase psychological switching costs** and directly influence Indonesian users' intention to stay. Conversely, **poorly targeted loyalty incentives** lead to budget waste by rewarding already-loyal users while ignoring those at risk of leaving.

Data analytics is vital to optimize marketing efficiency, as retaining users is significantly more cost-effective than new acquisition. Combining transaction-based behavioral segmentation (RFM) with explainable churn prediction gives the business team actionable insights to design precise, timely, and targeted interventions.

This project **simulates a Research & Data Analytics** team's workflow in **supporting** a live **loyalty program**. It focuses on mapping behavioral segments, predicting 90-day churn risks, and delivering segment-specific, data-driven retention strategies to maximize budget efficiency.



Problem Questions



1.

What behavioral segments exist among users based on the dimensions of Recency, Frequency, Monetary, and Engagement?



2.

Which model yields the highest churn prediction accuracy, and which features are the most predictive?



3.

How does churn probability differ across segments and tiers, and how much revenue is at risk per segment?

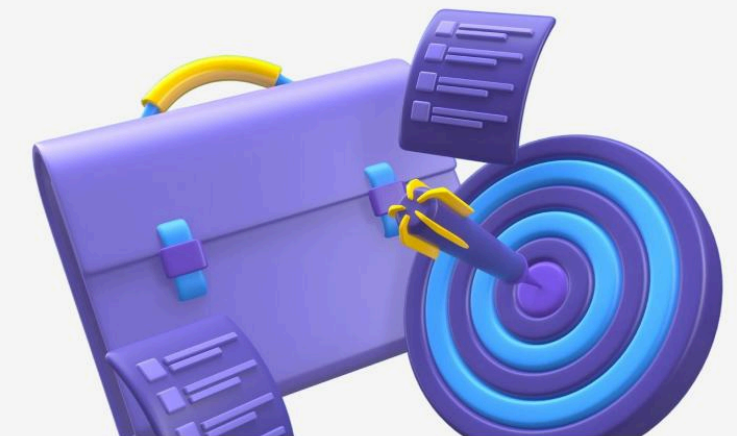


4.

Which retention interventions deliver the highest estimated ROI per segment?



Objectives



1.

To identify and map user behavioral segments by evaluating historical usage patterns across the dimensions of Recency, Frequency, Monetary, and Engagement (RFME).



2.

To develop and evaluate multiple machine learning models to determine which yields the highest accuracy for churn prediction, while pinpointing the most predictive underlying features.



3.

To quantify and compare churn probabilities across different user segments and membership tiers, and estimate the total revenue at risk within each identified segment.



4.

To evaluate and recommend targeted retention interventions that deliver the maximum estimated Return on Investment (ROI) tailored specifically to each user segment.



Dataset

This dataset is a synthetic dataset designed to simulate real user behavior in a loyalty program. The distribution parameters for each segment are based on patterns published in loyalty program and customer churn research literature.

Number of users	10,000 row
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Number of columns	12
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Overall churn rate	34.4% (not 32% as in the initial design, due to realistic noise)
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Duplicates	0 (clean dataset)
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Missing values	0 (100% complete)
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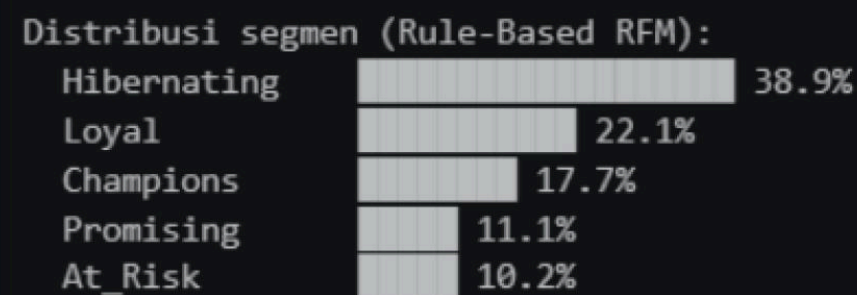
Reference date	January 1, 2025
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Format	CSV (rewards_synthetic.csv)
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Insight Analysis



1. Hibernating dominates the user base, not a small segment



The **Hibernating** segment is the **largest group** at **38.9%** of total users (rule-based RFM), far exceeding the initial design of 20%.

This indicates that the majority of the user base is already inactive and requires an immediate win-back strategy, not just maintenance.

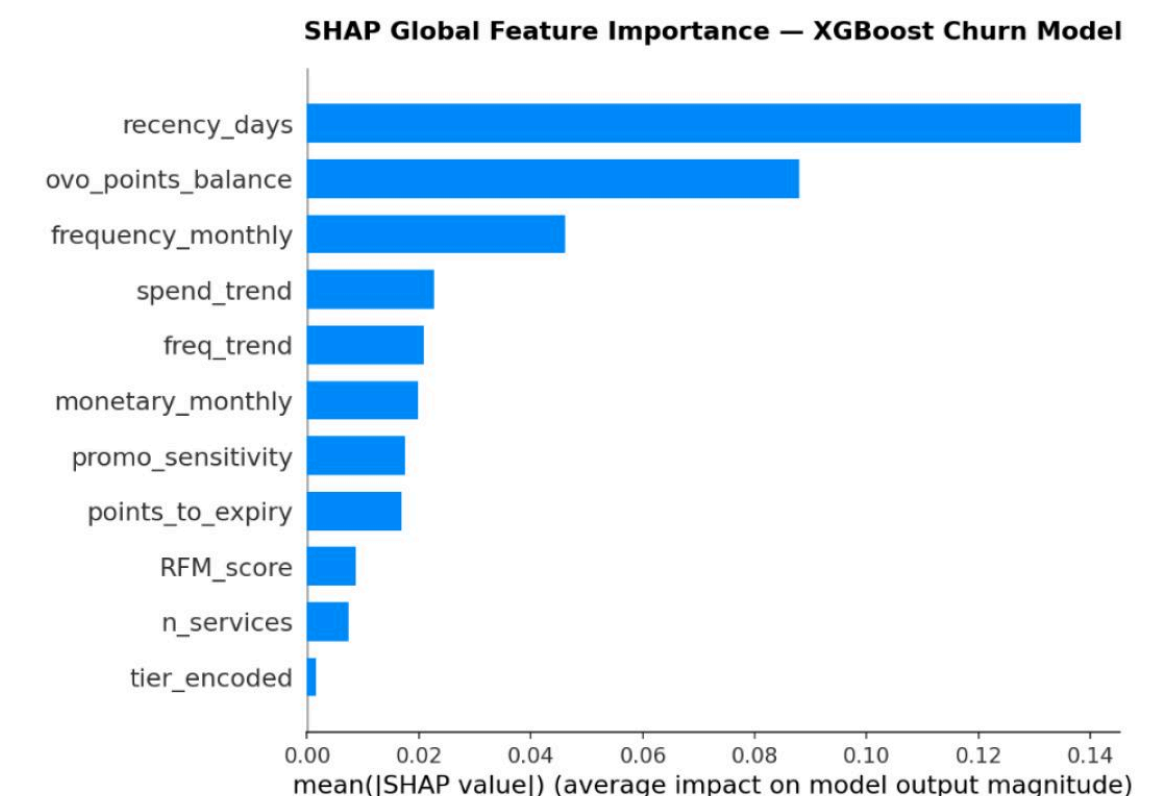
2. Churn rate is highly skewed across segments

	N Users	Avg Recency	Avg Freq/bln	Avg Monetary	Avg OVO Pts	Avg Services	Churn Rate
Segmen							
Champions	1,500	7.10	22.15	Rp 1,646,495	16441.13	2.69	3.0%
Loyal	2,500	18.26	10.93	Rp 549,638	5493.34	1.98	10.0%
At_Risk	2,000	66.69	3.48	Rp 227,739	2273.84	1.99	46.0%
Promising	2,000	11.73	5.00	Rp 165,935	1657.34	2.01	26.0%
Hibernating	2,000	226.70	0.00	Rp 40,460	405.36	2.01	85.0%

EDA analysis reveals extreme disparity across ground truth design segments: **Hibernating reaches 85%** churn rate, At_Risk 46%, while **Champions is only 4%** and Loyal 10%.

This confirms that one size fits all interventions are ineffective, each segment requires a different approach.

3. recency_days is the most dominant churn predictor



Based on SHAP analysis, **recency_days** is the feature with the **highest global** importance (mean $|SHAP| \approx 0.42$). For an example At-Risk user with a 70.9% churn probability, a recency of 49 days contributes **+0.15** to the churn prediction at the instance level, far above any other feature for that specific user.

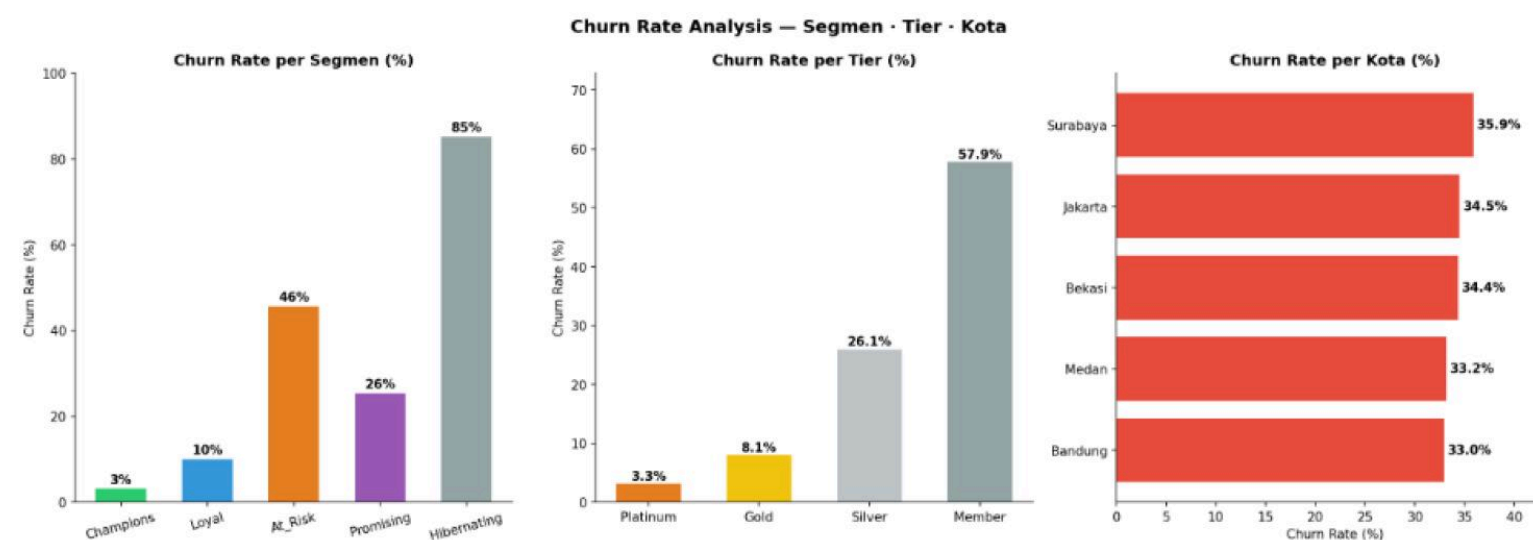
4. n_services is the second strongest predictor, ecosystem lock-in matters

Logistic Regression	76.8%	0.687	0.842	0.641	0.739
Random Forest	78.0%	0.694	0.839	0.664	0.728
XGBoost ✓	77.3%	0.689	0.838	0.651	0.731

Classification Report – XGBoost:				
	precision	recall	f1-score	support
Aktif	0.85	0.80	0.82	1313
Churn	0.65	0.73	0.69	687

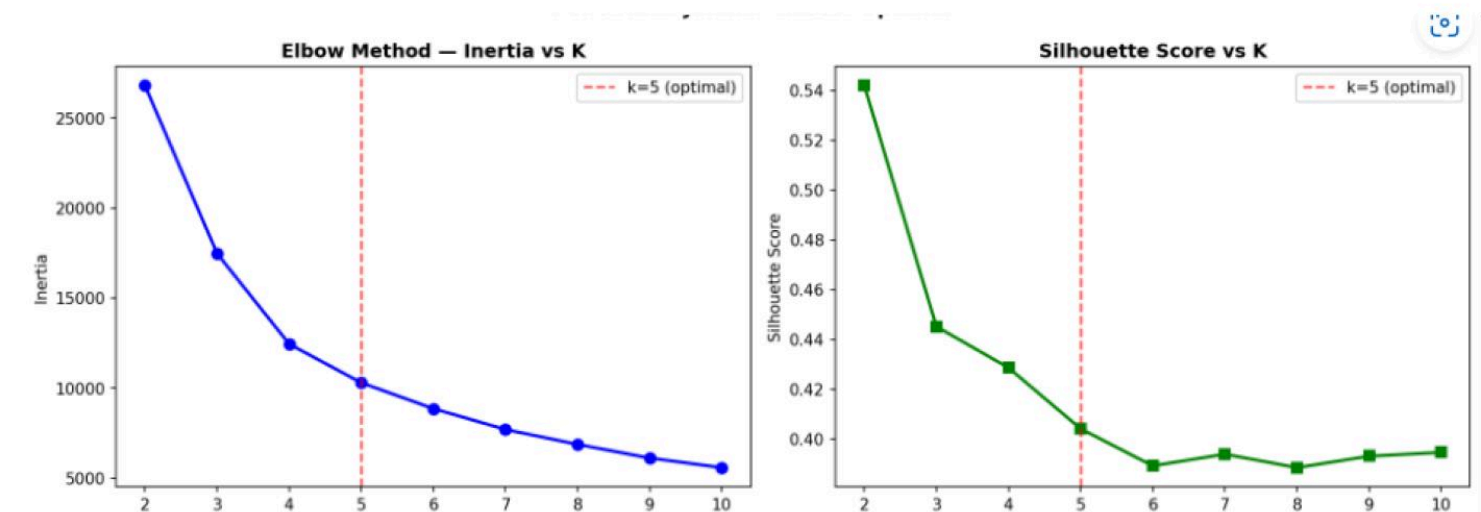
Based on SHAP analysis, n_services ranks #2 globally (mean |SHAP| $\approx$ 0.28). Users using only 1 service have a churn risk 3.2× higher than users with 3+ services. This makes cross-service adoption a critical early retention lever, especially for Promising segment users.

5. Member tier is most vulnerable to churn



Churn analysis per tier shows **Member tier has a 57.9%** churn rate, in stark contrast to **Platinum which is only 3.3%**. This inverse relationship between tier and churn confirms the importance of tier upgrade programs as a retention strategy.

6. K-Means detects 5 business-meaningful clusters



Final K-Means (k=5) yields **ARI 0.485**, moderate alignment with ground truth labels. Cluster 1 ("Champions-like") has a recency of 7 days, frequency of 22x/month, monetary ~Rp 1.7 million; Cluster 3 ("Lost Customers") has recency of 258 days and frequency of 0x/month, proving that clustering captures real behavioral patterns even without label supervision. Best silhouette score is 0.542 at k=2; k=5 was chosen for consistency with the 5-segment business framework.

7. ML models are competitive and consistent

Model	Accuracy	F1	AUC-ROC	Precision	Recall
Logistic Regression	76.8%	0.687	0.842	0.641	0.739
Random Forest	78.0%	0.694	0.839	0.664	0.728
XGBoost ✓	77.3%	0.689	0.838	0.651	0.731

Classification Report – XGBoost:				
	precision	recall	f1-score	support
Aktif	0.85	0.80	0.82	1313
Churn	0.65	0.73	0.69	687
accuracy			0.77	2000
macro avg	0.75	0.76	0.76	2000
weighted avg	0.78	0.77	0.78	2000

All three models show highly competitive and consistent performance: Logistic Regression (AUC 0.842), Random Forest (F1 0.694), XGBoost (AUC 0.838, F1 0.689). Random Forest has the highest accuracy (78.0%) while Logistic Regression leads in AUC-ROC (0.842). XGBoost was selected as the final model due to its best balance across all metrics, native scale_pos_weight support, and full SHAP TreeExplainer compatibility.

8. Revenue at risk reaches Rp 583 million/month. At_Risk is the highest-value target

Segmen	N Users	Churn%	Revenue at Risk	
Champions	1,500.0	3%	Rp	80,678,000
Loyal	2,500.0	10%	Rp	140,158,000
At_Risk	2,000.0	46%	Rp	208,837,000
Promising	2,000.0	26%	Rp	84,627,000
Hibernating	2,000.0	85%	Rp	69,024,000
TOTAL			Rp	583,324,000

Revenue at risk is calculated as $n_users \times avg_monetary \times churn_rate$ per segment. At_Risk contributes Rp 208 million/month, the largest single-segment risk despite being only 10.2% of users. Loyal segment follows at Rp 140 million/month. Hibernating, despite 85% churn rate, contributes only Rp 69 million due to near-zero monetary values.

Strategic Recommendations



Alert "OVO Points expiring in 30 days" + 25% off GrabFood voucher; trigger at recency > 45 days.



Win-back campaign only for users with historical monetary > Rp 500K (est. top 20% = ~400 users), skip low-value as voucher cost likely exceeds recovery value.



Push notification "X more points to reach Gold" + weekly frequency-based challenge; tier upgrade reduces churn from ~10% to ~3% (Silver → Gold).



"Try GrabMart, earn 2x points this week", push multi-service adoption; users with $n_{services} \geq 3$ have 3.2× lower churn risk.



Early access to new features + monthly double-points event + personal appreciation notification.

Conclusion



Maximizing Retention & Protecting Revenue

By leveraging an end-to-end data framework using K-Means clustering and an XGBoost predictive model (AUC-ROC: 0.838), this analysis successfully mapped 10,000 users and captured 73.1% of actual churn. Driven by key global predictors recency of transactions and ecosystem multi-service adoption the framework pinpoints the At_Risk segment as the highest-value target, carrying Rp 208 Million/month of the total Rp 583 Million monthly revenue at risk. Transitioning from a generic promotional strategy to automated, personalized micro-incentives (such as triggered food vouchers and cross-service adoption campaigns) allows the business to efficiently safeguard over Rp 150 Million+ per month in baseline revenue while drastically reducing voucher burn rate.



Any Questions?

Thank You

For Your Attention